MINIMAL OBJECTS OF THE CATEGORY OF MORPHISMS TO A FIXED OBJECT UNDER A BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction of a competing morphism in the target category to the source category through (S1) and the resulting identicalness of the image of the minimal object, and the lifting along a surjection with the control of the resulting object by minimality.

Editorial note. The label lemma-smallest-where-descends names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-smallest-where-descends –start 4321 –end 4372 –local-id 28021 returned RESULT=OK (L28021 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Construction of the pushforward of a morphism along a ring map. Same chapter, the construction of f_* of a morphism y -> x along a ring map f, using the (S1) property to produce the displayed square. Supplies the construction of f_* and the displayed diagram lying over B' -> f(B') inside B -> A.
- (S1) property of the category. Same chapter, the statement of (S1) as the existence of objects over a fibre product of Artinian quotients. Supplies the objects y" and y' used in the two parts of the proof.
- Existence of minimal objects dominated by a given one. Same chapter, the remark that every object of the category of morphisms admits a morphism from a minimal one, by boundedness of the lengths. Supplies the minimal object y" -> y used at the end of the second part.

Lemma 0.1. *Let $\mathcal{F}$ be a category cofibred in groupoids over $\mathcal{C}_\Lambda$ which has (S1).*

(1) *For $y \to x$ in $\mathcal{F}$ a minimal object in $\mathcal{S}_y$ maps to a minimal object of $\mathcal{S}_x$.*
(2) *For $y \to x$ in $\mathcal{F}$ lying over a surjection $f : B \to A$ in $\mathcal{C}_\Lambda$ every minimal object of $\mathcal{S}_x$ is the image of a minimal object of $\mathcal{S}_y$.*

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Proof. Proof of (1). Say $y \to x$ lies over $f : B \to A$. Let $y' \to y$ lying over $B' \subset B$ be a minimal object of $\mathcal{S}_y$. Let

$$\begin{array}{ccc} y' & \longrightarrow & x' \\ \downarrow & & \downarrow \\ y & \longrightarrow & x \end{array} \quad \text{lying over} \quad \begin{array}{ccc} B' & \longrightarrow & f(B') \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \end{array}$$

be as in the construction of f_* above. Suppose that $(x'' \to x) \to (x' \to x)$ is a morphism of $\mathcal{S}_x$ with $x'' \to x'$ lying over $A'' \subset f(B')$. By (S1) there exists $y'' \to y'$ lying over $B' \times_{f(B')} A'' \to B'$. Since $y' \to y$ is minimal we conclude that $B' \times_{f(B')} A'' \to B'$ is an isomorphism, which implies that $A'' = f(B')$, i.e., $x' \to x$ is minimal.

Proof of (2). Suppose $f : B \to A$ is surjective and $y \to x$ lies over f. Let $x' \to x$ be a minimal object of $\mathcal{S}_x$ lying over $A' \subset A$. By (S1) there exists $y' \to y$ lying over $B' = f^{-1}(A') = B \times_A A' \to B$ whose image in $\mathcal{S}_x$ is $x' \to x$. So $f_*(y' \to y) = x' \to x$. Choose a morphism $(y'' \to y) \to (y' \to y)$ in $\mathcal{S}_y$ with $y'' \to y$ a minimal object (this is possible by the remark on lengths above the lemma). Then $f_*(y'' \to y)$ is an object of $\mathcal{S}_x$ which maps to $x' \to x$ (by functoriality of f_*) hence is isomorphic to $x' \to x$ by minimality of $x' \to x$. $\square$